

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed systems at scale. Led infrastructure teams at **YouTube** and **Google**, shipping video streaming and search systems serving **10M+ concurrent users**. Expertise spans **Python, Go, distributed systems**, and **Kubernetes** across **GCP** and **AWS**. Scaled real-time marketplace infrastructure at **Whatnot** handling **50,000 concurrent users** during live events. Background includes **microservices architecture**, **search ranking systems**, and cloud cost optimization delivering **\$2M annual savings**. Partners with product and data teams to define technical strategy and drive end-to-end delivery.

Core Skills

Languages: Python, Go, C#, SQL, Bash, TypeScript, JavaScript, HTML, CSS

Distributed Systems: Microservices, High Availability, Load Balancing, Message Queues, Rate Limiting, Circuit Breakers

Infrastructure: Kubernetes, GCP, AWS, CI/CD, Terraform, Docker

Data: MySQL, Redis, PostgreSQL, Cassandra, Kafka, Memcached

Platforms: Video Streaming, Search Infrastructure, E-Commerce, Real-Time Systems

Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed distributed video delivery infrastructure processing millions of requests per second with p99 latency under **100ms** at **YouTube**.
- Led architecture reviews and technical decisions for video platform infrastructure supporting global user base across **GCP**.
- Owned reliability of **30+ microservices** handling video pipeline, achieving **99.99% uptime** while optimizing compute costs.
- Mentored 5 engineers on distributed systems design, code review standards, and incident response for on-call rotations.
- Collaborated with product teams to define roadmap priorities and translate requirements into technical specifications.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built live marketplace platform serving **50,000 concurrent users** during peak events with **real-time bidding** and inventory sync.
- Architected **Python microservices** on **Kubernetes** handling end-to-end transaction flows from checkout to fulfillment.
- Reduced cloud spend by **\$2M annually** through resource right-sizing and spot instance utilization for batch workloads.
- Established monitoring and alerting standards using **Prometheus** and **Grafana**, cutting mean time to detection by **40%**.
- Participated in on-call rotation for production incidents, maintaining platform stability during high-traffic drops.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led **YouTube Search infrastructure** team serving **3 billion queries per day** with p99 latency under **100ms**.
- Managed team of **8 engineers**, conducting performance reviews, hiring interviews, and career development planning.
- Migrated **200+ services** to **microservices architecture** on **Borg**, improving deployment frequency and system reliability.

- Drove cross-functional initiatives to reduce infrastructure costs by **\$5M** through **memcache** optimization and query caching.

Software Engineer | Google | 02/2011 - 02/2015

- Developed ranking algorithms for **YouTube search** backend systems, increasing video click-through rates by **25%**.
- Optimized **C++** backend services and **MySQL** queries, reducing backend latency by **30%** for search requests.
- Partnered with search quality team to implement ranking features, improving user engagement metrics across the platform.
- Owned on-call responsibilities for search serving infrastructure, resolving P1 incidents and improving runbooks.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and implemented backend services for **Windows Update** handling distribution to **400M+ devices** worldwide.
- Built **C#** and **.NET** services on **Azure** cloud platform during early adoption phase, contributing to platform reliability.
- Automated deployment pipelines and testing frameworks, reducing release cycle time from weeks to days.
- Collaborated with kernel-level teams to optimize update delivery performance and reduce bandwidth consumption.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed test automation frameworks for **Windows Vista** kernel components using **C#** and internal testing tools.
- Executed integration tests for system services, identifying and documenting bugs for engineering resolution.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007